

Rules of Differentiation and Their Use in Comparative Statics

ECON 320 - Introduction to Mathematical Economics

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1 Rules of Differentiation for a Function of One Variable

Constant-Function Rule

Statement. If $y = k$ (constant), then $\frac{dy}{dx} = 0$.

Why? A flat line has zero slope: changing x leaves y unchanged.

Power-Function Rule

Statement. For $y = x^n$ with $n \in \mathbb{R}$, $\frac{dy}{dx} = nx^{n-1}$ whenever the derivative exists.

Power-Function Rule (Generalized)

If $y = cx^n$ with constant c , then $\frac{dy}{dx} = cnx^{n-1}$. More generally, for $y = [g(x)]^n$ with $n \in \mathbb{R}$, use the chain rule (Section 7.3) to obtain

$$\frac{dy}{dx} = n[g(x)]^{n-1}g'(x).$$

Economic example (marginal cost from a polynomial total cost). Let $C(Q) = Q^3 - 4Q^2 + 10Q + 75$. Then

$$MC(Q) = \frac{dC}{dQ} = 3Q^2 - 8Q + 10.$$

Interpretation: MC is increasing when Q is sufficiently large because the $3Q^2$ term dominates, capturing congestion/capacity effects.

2 Rules Involving Two or More Functions of the Same Variable

Sum–Difference Rule

If $y = f(x) \pm g(x)$, then $y' = f'(x) \pm g'(x)$.

Intuition. Slopes add: the small change in $f + g$ over Δx is the sum of the small changes in f and g .

Product Rule

If $y = f(x)g(x)$, then

$$y' = f'(x)g(x) + f(x)g'(x).$$

Why? Over Δx , the product changes both because f changes while g is held at its old level and because g changes while f is held at its new level.

Finding Marginal-Revenue (MR) from Average-Revenue (AR)

Let $AR(Q) = P(Q)$ be price as a function of quantity. Total revenue is $TR(Q) = P(Q) \cdot Q$. Then by the product rule,

$$MR(Q) = \frac{dTR}{dQ} = P(Q) + QP'(Q).$$

Linear-demand case. If $P(Q) = a - bQ$, then $MR(Q) = a - 2bQ$ —twice as steep as AR, crossing the quantity axis at half the intercept.¹

¹This appears in many texts; see also Varian's *Intermediate Microeconomics* for geometric intuition.

Quotient Rule

If $y = \frac{f(x)}{g(x)}$ with $g(x) \neq 0$, then

$$y' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}.$$

Relationship Between Marginal-Cost and Average-Cost

Average cost $AC(Q) = \frac{C(Q)}{Q}$ (for $Q > 0$). Differentiate using the quotient rule:

$$\frac{dAC}{dQ} = \frac{Q \cdot MC(Q) - C(Q)}{Q^2} = \frac{MC(Q) - AC(Q)}{Q}.$$

Key implication. AC is minimized where $MC = AC$. If $MC < AC$, AC is falling (the marginal “pulls” the average down), and if $MC > AC$, AC is rising.

Worked example. Suppose $C(Q) = 100 + 20Q + Q^2$. Then $MC = 20 + 2Q$, $AC = 100/Q + 20 + Q$. Setting $MC = AC$:

$$20 + 2Q = 100/Q + 20 + Q \Rightarrow Q^2 = 100 \Rightarrow Q^* = 10.$$

At Q^* , AC attains its minimum.

3 Rules Involving Functions of Different Variables

Chain Rule (single-variable composition)

If $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x)) g'(x).$$

Economics example (tax pass-through with compositional dependence). Suppose price P responds to an ad-valorem tax τ via $P = P^*(1 + \tau)$, and demand is $Q = D(P)$; consumer surplus depends on $Q(\tau) = D(P^*(1 + \tau))$. Then

$$\frac{dQ}{d\tau} = D'(P^*(1 + \tau)) \cdot P^*,$$

linking demand slope at the taxed price to pass-through.

Chain Rule (multivariable form / total differential preview)

If $y = f(u, v)$ with $u = u(x)$ and $v = v(x)$, then

$$\frac{dy}{dx} = \frac{\partial f}{\partial u} \frac{du}{dx} + \frac{\partial f}{\partial v} \frac{dv}{dx}.$$

This is the one-dimensional version of the multivariate chain rule and equals the total differential divided by dx .

Inverse-Function Rule

If $y = f(x)$ is differentiable, strictly monotone, and hence invertible with inverse $x = f^{-1}(y)$, then

$$\frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{f'(x)} \quad (\text{evaluate at the corresponding point}).$$

Use. Switching comparative-statics perspectives: sometimes $\frac{dP}{dQ}$ is easier to compute than $\frac{dQ}{dP}$; invert at the end.

Worked example (supply elasticity via inverse rule). If $Q_s = \alpha P^\eta$, then $\frac{dQ_s}{dP} = \alpha\eta P^{\eta-1}$. The inverse supply $P(Q) = \left(\frac{Q}{\alpha}\right)^{1/\eta}$ has

$$\frac{dP}{dQ} = \frac{1}{\alpha^{1/\eta}\eta} Q^{1/\eta-1} = \frac{1}{\frac{dQ}{dP}} \quad (\text{at corresponding } P, Q).$$

4 Partial Differentiation

Partial Derivatives

For $z = f(x, y)$, the *partial derivative* with respect to x is

$$\frac{\partial z}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h},$$

holding y constant; $\frac{\partial z}{\partial y}$ is analogous.

Economic interpretation. $\frac{\partial z}{\partial x}$ is the marginal effect of x on z when other determinants are held fixed (*ceteris paribus*).

Techniques of Partial Differentiation

All single-variable rules apply with the other variables treated as constants. For example,

$$\frac{\partial}{\partial x} [x^a y^b] = ax^{a-1}y^b, \quad \frac{\partial}{\partial y} [\ln(x+y)] = \frac{1}{x+y}.$$

Cross-partials. $\frac{\partial^2 f}{\partial x \partial y}$ often equals $\frac{\partial^2 f}{\partial y \partial x}$ when f is sufficiently smooth (Clairaut's theorem).

Geometric Interpretation

For $z = f(x, y)$, partial derivatives are the slopes of the tangent plane in the x - and y -directions:

$$dz \approx \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

Gradient Vector

The gradient $\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$ points in the direction of the steepest ascent; its magnitude is the rate of increase per unit step. In constrained optimization (later chapters), gradients are central to the Lagrange multiplier method.

Worked example (Cobb–Douglas marginal products). Let $Q = AK^\alpha L^\beta$. Then

$$\frac{\partial Q}{\partial K} = \alpha AK^{\alpha-1} L^\beta, \quad \frac{\partial Q}{\partial L} = \beta AK^\alpha L^{\beta-1}.$$

Diminishing marginal products arise when $\alpha, \beta \in (0, 1)$.

5 Applications to Comparative-Static Analysis

Market Model

Consider linear demand and supply with shifters:

$$\begin{aligned} Q^d &= a - bP + \gamma Y, \\ Q^s &= c + dP - \delta W, \end{aligned}$$

where Y (income) shifts demand and W (wage) shifts supply. Equilibrium $Q^d = Q^s$ implies

$$(a - c) + \gamma Y + \delta W = (b + d)P \quad \Rightarrow \quad P(Y, W) = \frac{a - c}{b + d} + \frac{\gamma}{b + d}Y + \frac{\delta}{b + d}W.$$

Comparative statics.

$$\frac{\partial P}{\partial Y} = \frac{\gamma}{b + d} > 0, \quad \frac{\partial P}{\partial W} = \frac{\delta}{b + d} > 0.$$

Interpretation. Positive demand (income) and cost (wage) shocks raise equilibrium price; the adjustment is dampened by the combined slope $b + d$.

National-Income (Keynesian Cross) Model

Let $Y = C + I + G$, with $C = c_0 + c_1(Y - T)$ and $T = \tau_0 + \tau_1 Y$. Solving:

$$Y = \frac{1}{1 - c_1(1 - \tau_1)} [c_0 - c_1\tau_0 + I + G].$$

Comparative statics.

$$\frac{\partial Y}{\partial G} = \frac{1}{1 - c_1(1 - \tau_1)} \quad (\text{fiscal multiplier}), \quad \frac{\partial Y}{\partial \tau_1} = \frac{c_1(Y - \text{autonomous part})}{[1 - c_1(1 - \tau_1)]^2} \quad (\text{sign depends on } Y \text{ level}).$$

In standard calibrations with $c_1 \in (0, 1)$, $\frac{\partial Y}{\partial G} > 0$ and increases with c_1 .

Input–Output Model

With technology matrix A and final demand d , equilibrium output is

$$x = (I - A)^{-1}d.$$

Comparative statics. A small change dd yields $dx = (I - A)^{-1}dd$, so the sensitivity of sector i to demand in sector j is the (i, j) -entry of the Leontief inverse. This maps neatly onto the chain rule idea: total effects combine direct and indirect requirements.

6 Note on Jacobian Determinants

Definition and Use

For transformations $\mathbf{y} = \mathbf{f}(\mathbf{x})$ with $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$, the *Jacobian* is

$$J(\mathbf{f}) = \det \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \dots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \dots & \frac{\partial y_n}{\partial x_n} \end{bmatrix}.$$

If $J \neq 0$ at a point, the inverse-function theorem guarantees a locally unique inverse mapping; in comparative statics, this ensures that small parameter changes map to well-defined changes in endogenous variables.

Two-Equation Linear System (signs via Cramer's Rule)

Let

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = u_1, \\ a_{21}x_1 + a_{22}x_2 = u_2. \end{cases}$$

Then $J = a_{11}a_{22} - a_{12}a_{21}$. If $J \neq 0$, the solution exists and is unique. The sign of $\frac{\partial x_1}{\partial u_1}$ equals $\frac{a_{22}}{J}$ etc., a compact way to determine the direction of comparative-static effects without fully solving.

Worked example (tax–subsidy incidence in linear model). With demand $Q^d = \alpha - \beta P$ and supply $Q^s = \gamma + \delta(P - s)$ (unit subsidy s), equilibrium satisfies

$$\begin{cases} Q = \alpha - \beta P, \\ Q = \gamma + \delta(P - s). \end{cases}$$

Write as $A\mathbf{z} = \mathbf{u}$ with $\mathbf{z} = (Q, P)$ and $\mathbf{u} = (\alpha, \gamma + \delta s)$. Then

$$A = \begin{bmatrix} 1 & \beta \\ 1 & -\delta \end{bmatrix}, \quad J = \det A = -(\beta + \delta) \neq 0.$$

Hence $\frac{\partial P}{\partial s} = \frac{\partial P}{\partial u_2} \frac{\partial u_2}{\partial s} = \frac{1}{J} \cdot \delta \cdot (-1) = \frac{\delta}{\beta + \delta} > 0$ (price rises with a producer subsidy), and $\frac{\partial Q}{\partial s} = \frac{\beta \delta}{\beta + \delta} > 0$.

Summary

- **Single-variable rules:** constants, powers (and generalized), sums, products, quotients, chain rule, inverse rule.
- **Economic links:** $MR = P + QP'$, $AC' = (MC - AC)/Q$.
- **Partial derivatives:** ceteris paribus effects; cross-partial and smoothness; gradient gives direction of steepest ascent.
- **Comparative statics:** solve (or use Cramer's Rule/Jacobians) to get signs and magnitudes of ∂ effects.
- **Jacobian:** nonzero $\Rightarrow$ local uniqueness and well-defined responses to shocks.

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